Linux and the shell

Software Management & Development · Session 2



IE University · MFT & MCSDI

Installs to start now

They download while we talk.

macOS

- Terminal → paste the command from brew.sh. It asks for your password and may install Apple's developer tools: let it run
- Meanwhile, download Docker Desktop (Apple Silicon) from docker.com

Windows

- Terminal as administrator → `wsl --install` (WSL already there with another distribution? `wsl --install -d Ubuntu`)
- Restart when it finishes, then install Docker Desktop

> Docker needs ≥ 20 GB free and macOS 15 or later. Not your case? Skip Docker today and tell me.

Why use the shell

How much did we spend on restaurants last quarter, and where?

```
grep ",restaurants," transactions.csv | cut -d, -f3 | sort | uniq -c | sort -rn
```

```
   5 Rodilla
   4 La Tagliatella
   2 Five Guys
```

The command is the record of the analysis: save it, share it, run it next quarter, run it on a server.

> Servers, containers and the cloud are operated from a shell. Often it's the only interface they have.

Terminal, shell and other words

The terminal is the window; the shell is the program inside it that runs your commands.

Word	What it is	Examples
Terminal	the window where you type and read	macOS Terminal, Windows Terminal, VS Code
Shell	the program that runs your commands	zsh, bash, PowerShell
Command line, CLI	working by typing commands instead of clicking	`git`, `docker`, `uv`
Prompt	what the shell shows when it waits for you	`ana@laptop ~ %`
Script	a file of commands run one after another	`balance.sh`

Small tools, combined

- Each program does one thing well
- Programs work together: the output of one can be the input of the next
- Text is the universal interface

The restaurants pipeline, one program at a time:

```
grep ",restaurants," transactions.csv | cut -d, -f3 | sort | uniq -c | sort -rn
```

`grep ",restaurants,"`	keeps the restaurant rows
`cut -d, -f3`	keeps the third column: the name
`sort`	puts equal names together
`uniq -c`	counts each run of equal names
`sort -rn`	orders by that count, largest first

The shell and the filesystem

Parts of a command

`ls` `-l -h` `/etc`

command options argument

- Options change the behaviour: `-lh` = `-l -h`; long options use `--all`
- Spaces separate arguments: quote `"my file.csv"`
- Every command returns an exit code: `0` = success. `echo $?` shows the last one

```
ls -lh --all ~/smd          # two short options and a long one
cp "my file.csv" backup/    # two arguments: the quotes keep the first one whole
ls nofile; echo $?          # not 0: the command failed
```

The filesystem tree

`/`	the root
`/home/<user>`	your files (macOS: `/Users`)
`/etc`	system configuration
`/usr/bin`, `/bin`	programs
`/usr/local/bin`	programs installed by hand
`/var/log`	logs
`/tmp`	temporary files

- Absolute path: from the root, `/home/ana/smd/ledger`. Works from anywhere
- Relative path: from the current directory, `ledger/data`. Depends on where you are
- `~` is your home folder: `~/smd` = `/home/ana/smd`

Navigating folders

`pwd`	where am I
`cd ledger`	go to a folder
`cd ..`	one level up
`cd ~`	back to your home folder
`ls -la`	list, with hidden files
`./balance.sh`	a file in this folder (`.` = here)

```
$ pwd
/home/ana
$ cd smd/ledger
$ ls
data
$ cd ..          # back to /home/ana/smd
```

Working with files

`mkdir -p a/b`	create folders
`cp`, `mv`	copy, move or rename
`rm`, `rm -r`	delete, for good
`cat`, `less`	print, page through
`head`, `tail`	first and last lines

```
cp data/transactions.csv data/backup.csv
head -n 3 data/transactions.csv
```

> There is no recycle bin.
>
> Read every `rm` twice.
>
> Never `rm -rf` a path you haven't checked.

Keyboard shortcuts

Tab	complete names
↑	previous commands
↓	next commands
Ctrl+R	search the history
Ctrl+C	stop the command
Ctrl+L	clear the screen
`man cmd`	the command's manual (q to quit)
`cmd --help`	a shorter summary of the options

> Press Tab often: if it doesn't complete, the name is wrong or the file isn't where you think.

stdin, stdout and stderr

Every program starts with three channels open, together called standard I/O (stdio). The shell decides where each one goes.

Stream	Number	By default	Used for
stdin	0	the keyboard	what the program reads
stdout	1	the terminal	its results
stderr	2	the terminal	errors and warnings

```
ls data nofile > list.txt     # the error still appears on screen
```

> Errors have their own stream so they don't end up mixed with your data, and you still see them when the results go to a file.

Redirection and pipes

`cmd > f`	stdout to `f`, replacing it
`cmd >> f`	append to `f`
`cmd < f`	stdin from `f`
`cmd 2> f`	stderr (stream 2) to `f`
`a | b`	stdout of `a` → stdin of `b`

```
grep groceries data/transactions.csv > groceries.csv
echo "checked $(date)" >> report.txt
wc -l < groceries.csv                  # 12
ls nofile 2> errors.txt
```

Text-processing tools

`grep -i -v -c`	filter lines
`cut -d, -f3`	one column
`sort -rn`	sort
`uniq -c`	count adjacent repeats
`wc -l`	count lines
`awk -F,`	columns, filters, sums

```
cut -d, -f4 data/transactions.csv | tail -n +2 | sort | uniq -c | sort -rn | head -n 3
```

```
  12 groceries
  11 restaurants
   9 transport
```

> `sort | uniq -c | sort -rn | head`: “what is most frequent?” for any column of any file.

Users, packages and the environment

File permissions

```
$ ls -l ~/smd/ledger
drwxr-xr-x  2 ana  ana   4096 Jan 12 10:02 data
-rwxr-xr-x  1 ana  ana    412 Jan 12 10:05 balance.sh
-rw-r--r--  1 ana  ana    180 Jan 12 10:07 report.txt
drwx------  2 ana  ana   4096 Jan 12 10:08 private
$ ls -l /etc/shadow
-rw-r-----  1 root shadow 1342 Jan  3 09:00 /etc/shadow
```

The first letter is the type (`d` folder, `-` file). Then three groups of three: owner, group, everybody else. On a file, r = read, w = write, x = run. On a folder, r = list it, w = add or delete files inside, x = enter it with `cd`.

Name	Type	Owner	Group	Others
`data`	`d` folder	`rwx` everything	`r-x` list, enter	`r-x` list, enter
`balance.sh`	`-` file	`rwx` read, write, run	`r-x` read, run	`r-x` read, run
`report.txt`	`-` file	`rw-` read, write	`r--` read	`r--` read
`private`	`d` folder	`rwx` everything	`---` nothing	`---` nothing
`/etc/shadow`	`-` file	`rw-` read, write	`r--` read	`---` nothing

Changing permissions and owners

`chmod +x balance.sh`	let everybody run it
`chmod go-r report.txt`	remove read for group and others (u owner, g group, o others, a all)
`chmod -R g+w data`	let the group write in the folder and everything inside it
`chmod 644 report.txt`	`rw-r--r--`: the usual for files
`chmod 755 balance.sh`	`rwxr-xr-x`: the usual for scripts and folders
`chmod 600 key.pem`	`rw-------`: only the owner (keys, passwords)
`chmod 700 private`	`rwx------`: only the owner can enter
`sudo chown ana report.txt`	make `ana` the owner
`sudo chown ana:staff report.txt`	change owner and group
`sudo chown -R ana:ana data`	the folder and everything inside it

Numbers: r = 4, w = 2, x = 1, added per group: 7 = `rwx`, 6 = `rw-`, 5 = `r-x`, 4 = `r--`, 0 = `---`. Giving files to another user takes `sudo`.

root and sudo

- root can do anything
- `sudo cmd` runs one command as root, with your password, and logs it
- Least privilege: work as a normal user

```
$ apt install tree
E: Could not open lock file /var/lib/dpkg/lock-frontend - open (13: Permission denied)
$ sudo apt install tree
[sudo] password for ana:
```

> `sudo` and `chmod 777` are not fixes. Find out why it failed first.

Package managers

System	Tool	Install	Update all
Ubuntu / WSL	`apt`	`sudo apt install tree`	`sudo apt update && sudo apt upgrade`
macOS	`brew`	`brew install tree`	`brew update && brew upgrade`
Windows	`winget`	`winget install Git.Git`	`winget upgrade --all`

- Trusted repositories, dependencies resolved, repeatable: a list of packages is a reproducible install
- `curl … | sh` runs code you haven't read: only from official sources

```
$ brew install tree          # or: sudo apt install tree
$ which tree
/opt/homebrew/bin/tree       # Ubuntu: /usr/bin/tree
```

Environment variables

Named text values that every process receives and passes on to the programs it starts.

`HOME`	your home folder: `/home/ana`
`USER`	your user name
`SHELL`	your shell: `/bin/zsh`, `/bin/bash`
`PATH`	where commands are found (next slide)

```
echo $HOME
LEDGER_FILE=x.csv ./balance.sh  # one command
export LEDGER_FILE=x.csv        # this session
# permanent: add the export to ~/.zshrc
```

> No spaces around `=`. Without `export`, programs don't see it.

PATH: where commands are found

A list of folders. The shell searches them in order and runs the first match:

```
$ echo $PATH | tr ":" "\n"
/opt/homebrew/bin
/usr/local/bin
/usr/bin
/bin
$ which python3
/opt/homebrew/bin/python3
```

- `command not found` → folder not in `PATH` (or open a new terminal)
- wrong version runs → another one comes first
- `./script.sh`: the current folder isn't in `PATH`

Setting up the course environment

Setup options: three tracks

Track	What you get	Who it suits
A. Native	your OS's terminal, package manager and Docker Desktop; on Windows, a real Ubuntu in WSL	≥ 16 GB, admin rights
B. Ubuntu VM	Multipass VM: `multipass launch 24.04 --name smd` + Docker inside	full isolation; native Docker not possible
C. Cloud	GitHub Codespaces: Linux + VS Code in the browser	managed or low-spec laptops

> Same commands on every track: you always work in a Linux-style shell. On Windows, keep projects in `~` inside Ubuntu, not in `/mnt/c`.

WSL: Linux inside Windows

The Windows Subsystem for Linux runs Linux programs on Windows, without dual boot and without managing a VM yourself.

- WSL 2 is a real Linux kernel in a lightweight VM that Windows starts and stops for you
- On top, a distribution: Ubuntu, with bash, `apt` and the usual tools
- Docker Desktop runs its Linux containers in the same VM

Integrated with Windows

- Ubuntu is a tab in Windows Terminal
- Linux files appear in File Explorer, under Linux
- The Windows disk is at `/mnt/c` from Linux, but it's slow: work in `~`
- VS Code opens Linux folders

Checking the installation

```
echo $SHELL
uname -sm
curl --version | head -n 1
brew --version      # macOS
apt --version       # Ubuntu (WSL)
docker run --rm hello-world
```

- `brew: command not found` → run the two “Next steps” commands Homebrew printed
- `Cannot connect to the Docker daemon` → start Docker Desktop; `permission denied … docker socket` → log out and in
- Windows: open the Ubuntu terminal, not PowerShell; projects in `~`, not `/mnt/c`